BASE24®

LinkX Product Summary

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SECTION 1: PREFACE

1.1 DOCUMENT CONTROL

Version	**Author**	**Date**
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SECTION 2: LINKX FEATURES

LinkX is used to link BASE24 and UNIX hosted applications – allowing transaction messages to pass back and forth between two different environments.

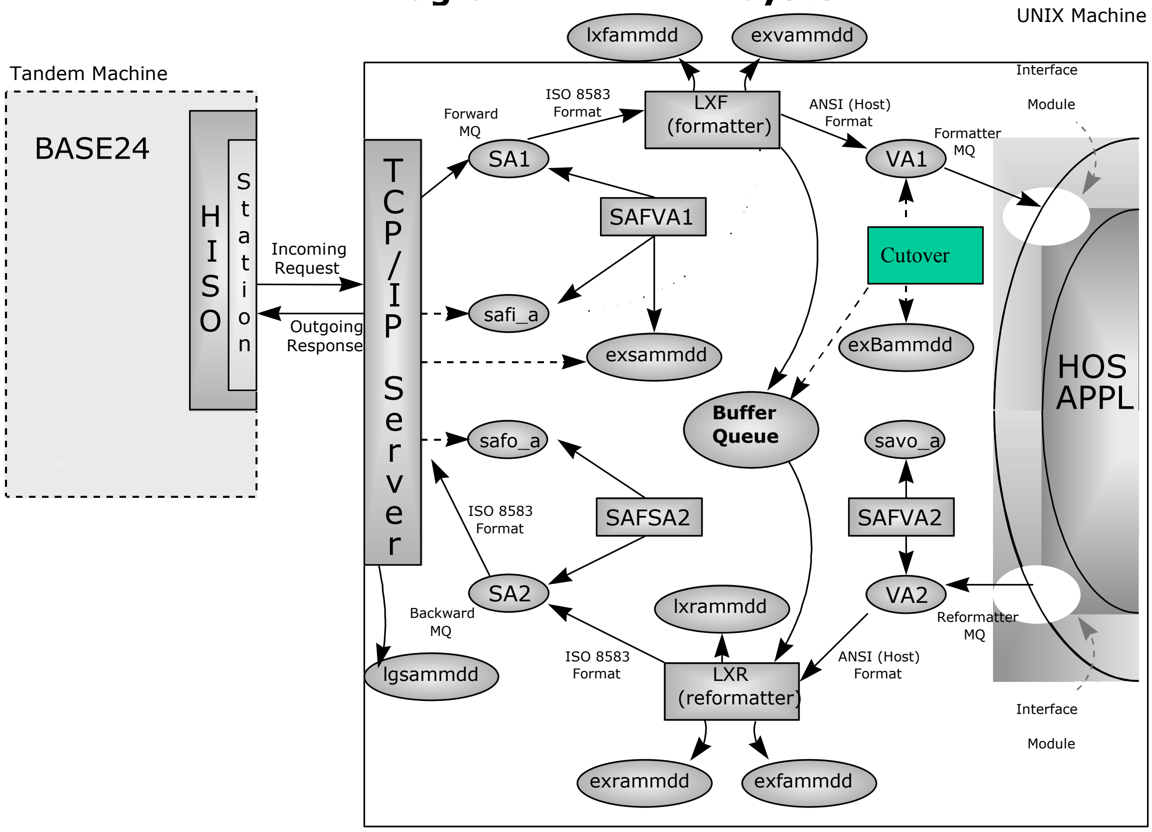
- Supports TCP/IP and X.25 protocol.
- Data Communication: LinkX takes full responsibility for data communications between BASE24 and UNIX platforms. It links user applications (host applications) on UNIX back-end hosts and BASE24/TANDEM by employing the reliable TCP stream sockets or X.25 PVC/SVC in full duplex mode.
- Supports LAN/WAN: LinkX supports local communications media like LAN as well as remote communications media over WAN or VSAT.
- Supports BASE24 Host Interface (ISO and ANSI), BASE24 Interchange Interface (ISO and ANSI).
- Message Formatting/Reformatting: LinkX supports ISO message format between BASE24 and hosts. To supports our customer's investments in current applications, LinkX can reformat messages to cater for the existing host applications' formats.
- Message Routing: LinkX can be deployed as a router and/or a gateway. LinkX can route request/response messages to/from different applications within the same UNIX host or different UNIX hosts.
- Mutli-channel: LinkX supports concurrent multi-sessions and multi-threaded communications, without blocking and waiting.
- Message Test Simulator: The simulator provides the ability to test message formatting/reformatting in a wide scope including BASE24-atm, BASE24-pos, VISA/MASTERCARD via BASE24-HISO, etc.
- Supports Acquirer/Issuer Transactions: LinkX supports both acquirer and issuer transaction processing for BASE24-atm, BASE24-pos, BASE24-teller, BASE24-telebanking, BASE24-fhm, BASE24-mail applications and so forth.
- Network Management: LinkX can handle network management (logon, logoff, echo test) between BASE24 and hosts.
- Store And Forward Files (SAF): LinkX can store messages into disk files at any sending points when the link between LinkX and BASE24 is down or when the queues are not available.
- Buffering messages: LinkX could pass selected fields instead of all fields in the message to the host applications and buffer the rest. When response message comes from host, LinkX could plug in buffered fields and send them back to BASE24.
- SAF/MQ monitor: SAF/Message Queue Monitor processes are daemon processes looking after pairs of SAF file and message queue. It is useful in the case of night-sleep host or limited system resources to ensure the working message-queues be available even for high traffic volume.
- Scalability: LinkX could be replicated within a machine or provide more routers to help improve traffic throughput.
- High-available environment: Upon request, LinkX could support high-available environment.

SECTION 3: LINKX SYSTEM DIAGRAMS

LinkX can be deployed into the following system architecture.

3.1 Single UNIX Host (with message formatter/reformatter):

Diagram 1. LINKX System



SECTION 4: LINKX ARCHITECTURE

LinkX consists of the following major processes/modules:

- LinkX Kernel(LX-Server, LX-Client),
- LinkX Message Formatter/Reformatter(LX-Formatter/Reformatter),
- LinkX Simulator(LX-Simulator),
- LinkX Real-time Monitor(LX-Monitor),
- LinkX Management Utilities(LX-Management).

4.1 LinkX Kernel

LinkX Kernel consists of LX-Server and LX-Client:

- LX-Server. This is a daemon process starting at the system-boot phase or by operators. It is a server process used to accept connection requests from BASE24 or other application hosts in gateway environment. After the connection between LinkX and BASE24 or between LinkX and hosts is established, LX-server interchanges messages between the two endpoints, logs all incoming and outgoing data, and provides a Store-and-Forward feature (SAF) together with the LinkX-formatter/reformatter or LinkX-Management. LX-server communicates with BASE24 or application hosts in the concurrent, full duplex way, without blocking. It routes the message to designated intra-machine or inter-machine applications via Standard UNIX Message-Queue based on the Product-id or processing-code contained in the BASE24 external-message. It can also route the message to another UNIX machine by using additional pair of LinkX client/servers on another port.

- LX-Client, This is used to communicate with LX-server on another UNIX host, it can route transaction messages to application systems via message queues or act as the a simulator to test communication with LX-server within the machine without BASE24. LX-client is a process running on demand. It also provides Save-And-Forward(SAF) feature.

4.2 LinkX Message Formatter/Reformatter

LinkX Message Formatter/Reformatter(LX-Formatter/Reformatter) is used when the UNIX host application uses a different format from BASE24 ISO 8583 format. As the format of UNIX host application differs from one customer to another and because the application environment is evolving, this component provides flexibility and meanwhile helps reduce the complexity of RPQ work.

LinkX Message Formatter/Reformatter consists of the following parts;

- Formatter: maps ISO 8583 message format into host applications desired format. LinkX provides all the fields information of the extern messages for customer to make further evolution.
- Reformatter: converts host application format back to ISO 8583 format. Customer can replace this module by his own coding.
- Mapping table.

As a software product, it should be as standard as possible and also meet most customers' requirements. However in reality, the host applications are different from one customer to another. Some customers have their own special proprietary requirements, e.g. they may only need a portion of a field, not the whole field; the field length of their own host message is not big enough to hold the whole field of BASE24 external message, but they want all the information of this field, and this whole field must be returned to BASE24, so this field in BASE24 external message must be moved to a "big" field of host message format for host application to use, and be moved back after host processed.

The LinkX-Formatter/Reformatter data flow charts are as follows.

Note: for simplicity sake, SAF feature is not shown in the following diagrams.

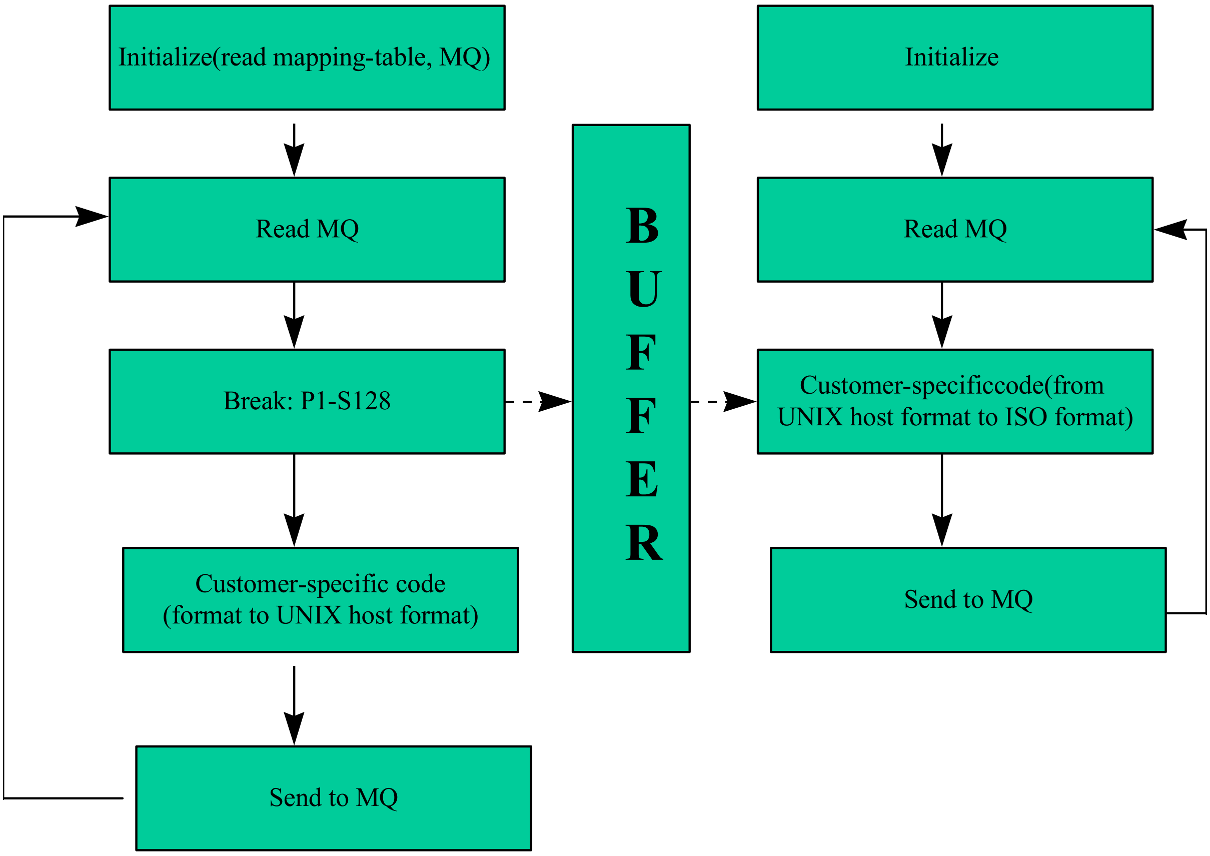


Figure 1: LX-Formatter data flow chart

Figure 2: LX-Reformatter data flow chart

4.3 LinkX Simulator

LinkX provides the UNIX host application simulator and BASE24 message simulator(LX-Simulator) to test the message formatting/reformatting. The simulator can simulate either the UNIX host application or BASE24 by different deployment.

UNIX Host applications are different from one customer to another, and messages to and from the UNIX host is dependent on UNIX host applications, so LinkX simulator should be flexible and provide an open interface allowing customer to plug in their special processing.

The data flow of application simulator is as follows.

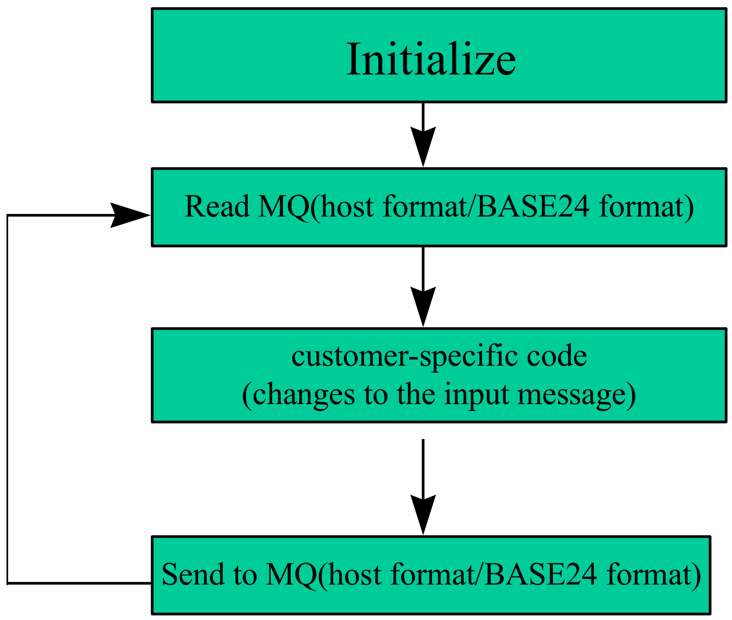


Figure 3. Application Simulator data flow chart

4.4 LinkX Real-time Monitor

LinkX Real-time monitor(LX-Monitor) consists of the following parts:

- LinkX System Monitor(auto-restart): LinkX provides facility to monitor the status of LinkX processes , message queues and SAF files at a given interval. When processes are detected to be down, it can restart the processes automatically.
- SAF/Message Queue Monitor: this daemon process looks after the SAF files and the message queues. It is exceptionally useful in the case of night-sleep host or limited system resources.

4.5 LinkX Management Utilities

LinkX Management Utilities(LX-Management) consists of the following parts:

- Start/Stop LinkX System: provides shell scripts to start or stop LinkX processes.
- On-line Config: provides online configuration modification capabilities for the operating parameters (warmbootable).
- Logging: logs all incoming and/or outgoing messages into disk files. All logged messages are marked with time stamp and “incoming” or “outgoing” headers.

Logon/Logoff/Echo-test: operators can issue the network management requests actively and manually.

SECTION 5: SYSTEM PRE-REQUISITES

LinkX is to install and work on a UNIX system with:

1. TCP/IP sockets library available for compilation and link or X.25 API available for compilation or link;
2. UNIX operating system which conforms to :
   IEEE POSIX std 1003.1 or later.
   X/Open Portability Guide, Issue 4 or later.
   AT&T SVID Issue2 or later;

3. ANSI-C conformable compiler and standard system libraries, a source-level debugger such as dbx;

4. TCP/IP subsystem properly configured and fully operational or X.25 subsystem properly configured and fully operational..

5. UNIX I/O, streams, message queue, I-node should be properly setup.
   Suggestion:
   a. Message Queue:
      Provided there are N transactions arrive to host during the longest processing cycle of the host application for a single most complex transaction, the maximum length of a transaction message is M bytes, thus the maximum length of a message queue must be greater than : (1.2 * N * M) bytes.

   b. Streams :
      Streams size should be greater than 1 M bytes.

   c. I-node:
      I-node should be greater than 100 system wide.

   d. Hard disk space
      Depending on the real transaction volume, for now reserve at least 10 mega bytes of hard disk space for LinkX every morning.

   e. TCP/IP ports
      It is recommended port 8270 for testing system, and prots from 8271 to 8279 used for production system. Reserve these ports for LinkX exclusively in every instance.

LinkX works with BASE24/XPNET 2.0, 2.1 and 3.0 only, so BASE24/XPNET must be correctly installed / configured and functioning in the Tandem.

A physical connection enabling TCP/IP or X.25 protocol between the UNIX machine and a Tandem machine must be available. The connection must be operational and can be tested using PING and FTP bidirectionally.

SECTION 6: PORTABILITY

LinkX is destined to run on a wide range of UNIX platforms, therefore it should be made portable with minimal modifications.

LinkX conforms to the following major UNIX standards and implementations:

IEEE POSIX 1003.1;
X/Open Portability Guide, Issue 4.
AT&T System V, issue 2.

Due to the heavy use of low-level I/O system calls by LinkX, and the differences in implementation and definitions among various UNIX vendors, a recompilation is typically needed.

SECTION 7: PACKAGE DELIVERABLES

There could be several types of LinkX packages deliverable from ACI depending on the customer's requirements.